

Low-Energy Electron Capture in Two-Dimensional Plasmonic Materials

A Dressed-Mass Mechanism, a Rigorous Rate Bound, and a Decisive Experimental Test

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Abstract

Converting a proton to a neutron by electron capture, $p + e^- \rightarrow n + \nu_e$, costs 0.782 MeV — roughly five orders of magnitude beyond any chemical, thermal, or condensed-matter energy scale. This single fact is why neutron-first proposals for low-energy nuclear reactions have remained contested. Strong-field quantum electrodynamics suggests one escape: an electron immersed in an intense coherent field acquires an effective rest mass $\tilde{m}c^2 = m_e c^2 \sqrt{1 + \xi^2}$, and the capture channel opens once the dimensionless field strength reaches $\xi \geq \xi_{\text{th}} \approx 2.32$. We develop this dressed-mass mechanism for a graphene/borophane geometry, in which terahertz acoustic plasmons confine the driving field most strongly, and then ask — rigorously — whether plasmon coherence can actually supply the 0.782 MeV deficit.

It cannot. The field must deliver the deficit in $n = \Delta E / \hbar \omega \approx 8 \times 10^7$ quanta, and the natural rescue — treating the plasmon as a coherent N-quantum object, whose required occupation appears to fall toward unity — fails for two independent reasons. First, the N-quantum coupling is not free: generated at N-th order it scales as $g_{\text{eff}} \sim g_0^N$, which exactly cancels the n^N coherence bonus and restores the multiphoton suppression $|J_N(\xi)|^2$. Second, the required occupation cannot fall below N at all, because a^N annihilates N quanta. The capture rate is therefore bounded roughly 10^6 orders of magnitude below detectability, even in the most favourable accessible case. This turns the rate from an open question into a rigorous upper bound — and turns the proposal's real content into an experiment. The downstream reaction $n + {}^{10}\text{B} \rightarrow {}^7\text{Li} + {}^4\text{He}$ provides a unique, artifact-resistant triple signature: a 478 keV gamma (the $\sim 94\%$ branch) in coincidence with ${}^4\text{He}$ and correlated heat, localized to plasmon hotspots and gate-tunable. The claim is deliberately modest: the mechanism is bounded by theory, and the question is decidable only by measurement.

***Authorship and status.** A research-stage proposal, derived from the author's prior work [1] and revised following detailed technical review. Portions were prepared with AI assistance under the author's direction. The author does not hold formal expertise in strong-field QED or nuclear physics, and the document has not undergone independent expert review. The central rate result (§3, Appendix B) is presented as a bound; the experimental program (§9) is the paper's operative contribution. Formal expert review, and the calculations identified in §10, are required before any practical interpretation.*

1. The Problem and the Proposal

The weak process $p + e^- \rightarrow n + \nu_e$ is endothermic by 0.782 MeV when both reactants are at rest. No chemical bond, thermal excitation, or condensed-matter degree of freedom comes within five orders of magnitude of closing that gap — the principal reason that claims of nuclear reactions on hydrogen-loaded surfaces have been met with skepticism for thirty years [2, 3]. One route sidesteps the Coulomb barrier of charged-particle fusion by routing the first nuclear step through neutron formation [7]: an ultra-low-momentum neutron,

once created, is absorbed by a neighbouring nucleus with no Coulomb obstruction at all. Everything then hinges on a single question — can the capture step itself proceed at a useful rate?

This paper develops the dressed-mass route to that capture step in a graphene/borophane geometry, and then settles its rate. It makes three changes to the Widom–Larsen picture and — unlike earlier treatments — establishes the rate as a rigorous bound rather than asserting it:

- **A two-dimensional geometry.** Graphene [8] supplies the conduction electrons and the hydride borophane [1, 9] supplies the protons; acoustic graphene plasmons (AGPs) confine terahertz fields far more tightly than any three-dimensional plasmon.
- **A downstream reaction with clean signatures.** The capture product — an ultra-low-momentum neutron — fuses with ^{10}B to give $^7\text{Li} + ^4\text{He}$, releasing charged-particle energy with no hard primary gamma.
- **A rigorous rate analysis.** Section 3 shows that neither raising a single electron's energy nor exploiting plasmon coherence lowers the threshold; the rate is bounded far below detection. The paper therefore leads with the experiment that can decide the question regardless of mechanism.

The logic of what follows is linear. Section 2 fixes the energy gap and introduces the dressed-mass mechanism that could, in principle, open the channel. Section 3 — the technical core — shows by two independent arguments why paying the 0.782 MeV with plasmon quanta fails, leaving a rate bounded roughly 10^6 orders of magnitude below detection. Sections 4 and 5 audit the two physical inputs that bound depends on: the field actually seen by an adsorbed proton, and what graphene plasmons can really deliver. Sections 6–9 then build the downstream reaction and the single coincidence experiment that settles the question whether or not the mechanism works, and Section 10 states precisely what is, and is not, being claimed.

2. The Energy Gap and the Dressed-Mass Mechanism

With PDG masses [3], the free-space capture Q-value is

$$Q_{\text{free}} = (m_p + m_e - m_n)c^2 = -0.782 \text{ MeV}. \quad (1)$$

A single electron moving at the graphene Fermi velocity carries only about 2.84 eV — smaller than the deficit by a factor of 275,000. The mechanism therefore cannot draw on electron kinetic energy. It relies instead on the intensity-dependent (Volkov) rest mass that a charge acquires in an intense oscillating field [4, 5]:

$$\tilde{m}c^2 = m_e c^2 \sqrt{1 + \xi^2}, \quad \xi \equiv eA_{\text{rms}}/(m_e c). \quad (2)$$

Capture becomes energetically allowed once the dressed mass reaches the neutron–proton splitting, $\tilde{m} \geq m_n - m_p$; equivalently $\xi \geq \xi_{\text{th}} = \sqrt{[(m_n - m_p)/m_e]^2 - 1} \approx 2.32$ (Appendix A). Figure 1 shows the crossing.

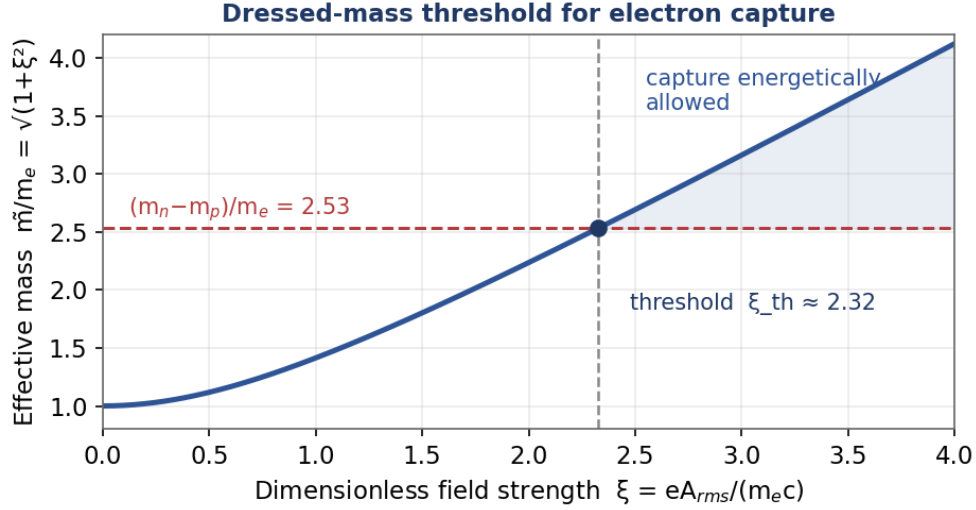


Figure 1. The effective rest mass grows with the dimensionless field strength ξ . Capture becomes energetically allowed once $\tilde{m}/m_e$ exceeds $(m_n - m_p)/m_e = 2.53$, which occurs at $\xi_{th} \approx 2.32$.

At 4 THz this threshold corresponds to $E_{rms} \approx 1.0 \times 10^{11} \text{ V} \cdot \text{m}^{-1}$ — at the extreme upper edge of reported graphene near-fields [10, 11]. (For calibration, the SLAC E144 experiment operated at $\xi \lesssim 0.4$ [6], well below this regime; it confirmed dressed-electron kinematics but not the conversion of field energy into rest mass for a weak decay.) Whether the field — or the plasmon's coherence — can actually supply the 0.782 MeV is the subject of §3.

3. The Rate: Why Coherent Plasmon Coupling Does Not Lower the Threshold

The dressed-mass threshold is real, but reaching it is an energetic statement, not a rate. The energy must be supplied to the weak vertex in quanta of the driving mode, and this section asks whether plasmon coherence can make that affordable. Two independent arguments — one dynamical, one purely kinematic — show that it cannot.

3.1 The multiphoton barrier

Because energy arrives in quanta of the drive, the process is intrinsically of very high order,

$$N = E_C / \hbar\omega, \quad E_C = 782 \text{ keV}. \quad (3)$$

At a representative AGP energy $\hbar\omega \approx 10 \text{ meV}$, this is $N \approx 7.8 \times 10^7$. A coherent (classical) field gives the standard Volkov amplitude for absorbing N quanta, the generalized Bessel function $J_N(\xi)$; for $N \gg \xi$ it is hyperexponentially small,

$$|J_N(\xi)|^2 \approx (2\pi N)^{-1} (e\xi/2N)^{2N}.$$

This is the barrier that any rescue must overcome.

3.2 The coherent N -quantum rescue, and why it cancels

The natural rescue treats the AGP as a coherent N -quantum object with an effective vertex,

$$H_{int} = g_{eff}[(a^\dagger)^N \sigma_- + a^N \sigma_+], \quad \Gamma \propto |g_{eff}|^2 \langle n^N \rangle \approx |g_{eff}|^2 n^N, \quad (4)$$

from which the required occupation $n^* \gtrsim (\gamma/g_{\text{eff}})^{2/N}$ appears to fall toward unity for large N . The error is that g_{eff} is not a free constant. The operator $(a^\dagger)^N$ is generated only at N -th order from the fundamental single-quantum coupling g_0 , so

$$g_{\text{eff}} \sim g_0^N / \Delta^{N-1}, \quad \Delta \sim \hbar \omega_0 \Rightarrow \Gamma \propto |g_0^N|^2 n^N = (g_0 \sqrt{n} / \Delta)^{2N} = \xi^{2N}. \quad (5)$$

In words: coherence does multiply the rate by the bonus n^N , but assembling an N -quantum vertex out of the one-quantum coupling costs exactly g_0^N ; and since $\xi \equiv g_0 \sqrt{n} / \Delta$, the two factors combine into ξ^{2N} — precisely the multiphoton suppression of §3.1, now viewed from the coherent-state side. The n^N bonus and the g_0^N penalty are the same physics counted once each. Keeping the bonus while treating g_{eff} as order-unity is exactly what produces the false conclusion; restoring the physical coupling returns $|J_N(\xi)|^2$ precisely. A macroscopically occupied plasmon is a coherent state, so this picture does not sit outside the multiphoton result — it reproduces it.

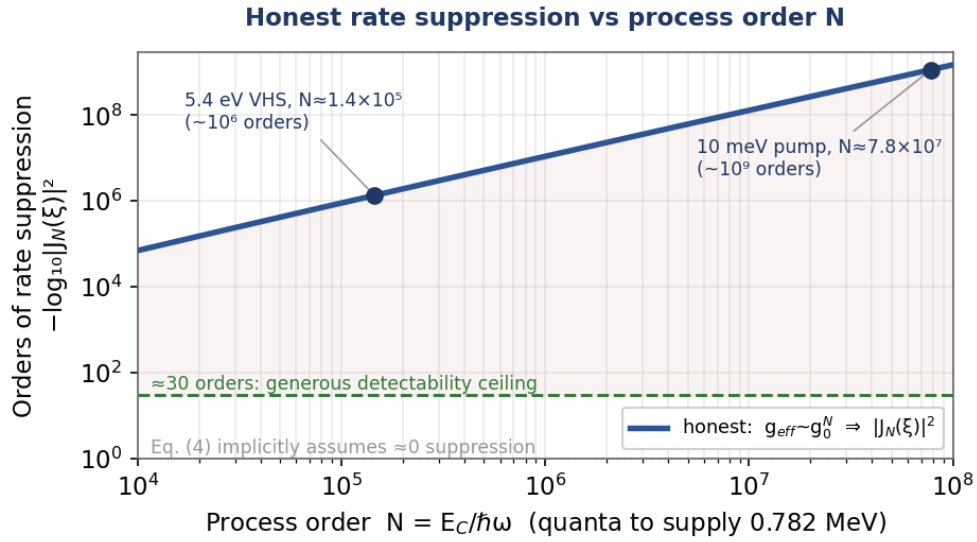


Figure 2. Honest rate suppression $-\log_{10}|J_N(\xi)|^2$ versus process order N , at $\xi = 2.3$. Even the most favourable accessible drive (a 5.4 eV Van Hove resonance, $N \approx 1.4 \times 10^5$) lies roughly 10^6 orders below any plausible detectability ceiling; an order-independent coupling, Eq. (4), would place this entire curve at zero.

3.3 The kinematic floor

A second obstruction, free of any dynamical assumption, confirms the result. Because $a^N|n\rangle = \sqrt{[n!/(n-N)!]}|n-N\rangle$ vanishes for $n < N$, the mode must contain at least N quanta before it can supply N . Any expression that returns a required occupation below N is therefore unphysical — yet the coherent-rescue formula does exactly that, giving $n^* \approx 4.3 \times 10^6$ while $N = 7.8 \times 10^7$. Enforcing $n \geq N$ removes the apparent advantage: the minimum occupation is N , and the rate still carries the $|J_N|^2$ suppression.

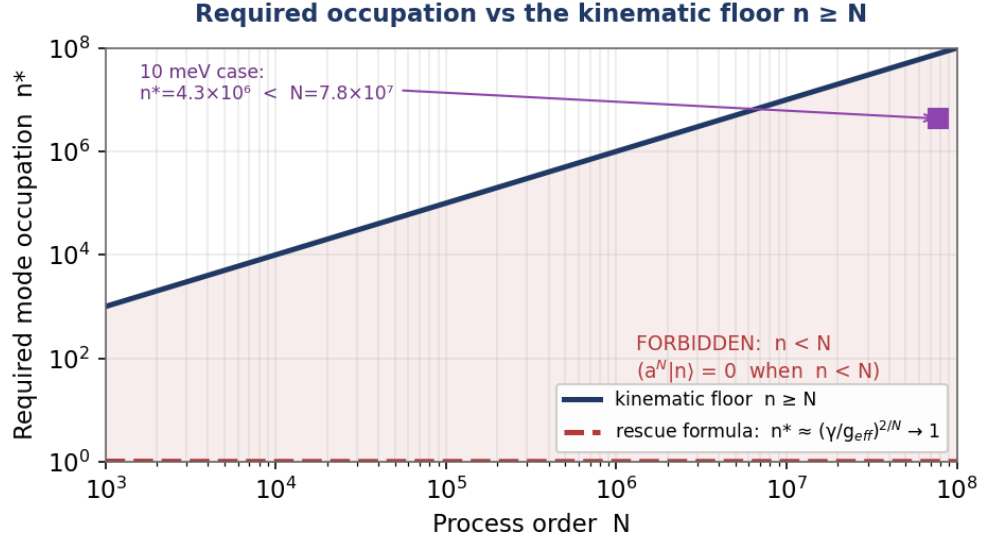


Figure 3. The coherent-rescue formula drives the required occupation n^* below the process order N , into the region forbidden by $a^N|n\rangle = 0$. The physical floor is $n \geq N$.

3.4 The bound

Evaluating the honest rate at $\xi \approx 2.3$ across the drives available in a graphene system gives Table 1.

Table 1. Only the single-quantum channel ($N = 1$, a 0.782 MeV γ -ray) is unsuppressed — and no plasmon or laser exists at that energy. Every coherent THz-to-optical drive is suppressed by 10^6 – 10^9 orders. The final column lists the (unphysical) value that follows from an order-independent coupling.

Coherent drive $\hbar\omega$	$N = Ec/\hbar\omega$	Honest $\log_{10} J_N(\xi) ^2$	Rescue-formula claim
10 meV (THz plasmon)	7.8×10^7	$\approx -1.2 \times 10^9$	$n^* \rightarrow 1$ (4.3×10^6)
0.20 eV (optical phonon)	3.9×10^6	$\approx -5 \times 10^7$	$n^* \rightarrow 1$
5.4 eV (M-point VHS)	1.4×10^5	$\approx -1.4 \times 10^6$	1.2×10^6
0.782 MeV (γ , $N=1$)	1	$O(1)$, unsuppressed	trivial

Conclusion. None of the available levers crosses zero. Confinement raises g_0 per quantum, but it enters the rate only through ξ , and pushing ξ to order unity is nothing other than the $\sim 10^{11} \text{ V} \cdot \text{m}^{-1}$ dressed-mass field by another name. The capture rate is bounded roughly 10^6 orders of magnitude below detection even in the best accessible case. Within single-particle and coherent-state physics the threshold is simply not lowered; a non-negligible rate would require a genuinely new, strongly-correlated many-body channel that is not established. We therefore treat the rate as an upper bound (Appendix B) and design the test of §9 so that it does not depend on it.

4. The Field an Adsorbed Proton Sees

The proton does not sit in the bare mode field but in a screened, geometrically reduced local field. Writing $A_{\text{loc}} = \eta_{\text{loc}} A_0$, the local-field factor η_{loc} is a product of evanescent decay, a bound-state form factor, RPA screening, and a Debye–Waller factor (Appendix C). At $q \approx 10^7 \text{ m}^{-1}$ and an adsorbate height of $\sim 1 \text{ \AA}$, the purely geometric reduction is small: $\eta_{\text{loc}} \approx 0.999$. Two caveats keep this from settling the matter. First, η_{loc}

as computed captures geometric decay only — not the strong-field, many-body screening of a longitudinal field at a charged adsorbate, which could be far larger. Second, the required $\sim 10^{11} \text{ V}\cdot\text{m}^{-1}$ approaches dielectric breakdown, and the highest reported amplitudes are sub-nanometre tip-peak values, not coherent fields filling a uniform mode volume. The local-field factor is necessary, but not sufficient, to retire the screening objection.

5. What Graphene Plasmons Deliver

5.1 Dispersion and confinement

Graphene plasmons disperse as $\omega \propto \sqrt{q}$ [12], so at 4 THz the wavevector is $q \approx 10^7 \text{ m}^{-1}$ — about 100 times the free-space value. This confinement is exactly what allows two-dimensional plasmons to carry such intense near-fields.

5.2 Field amplitude

Reported amplitudes span several decades [12, 13]: far-field values of $\sim 10^7$ – $10^9 \text{ V}\cdot\text{m}^{-1}$, tip-enhanced near-fields of 10^9 – 10^{10} , and the highest sub-nanometre tip values of 10^{10} – 10^{11} . The dressed-mass threshold sits at the very top of this range — and those peak values are not sustained over a uniform mode volume.

5.3 Coherence

Measured plasmon lifetimes are 0.1–1 ps [10, 11, 14], giving a single-cycle coherence factor $f_{\text{coh}} = \omega\tau/(1+\omega\tau)$ of 0.56–0.96 — close to unity over one cycle. But the N-quantum amplitude of §3 requires the drive to stay phase-coherent across all N absorption events, i.e. over $N \approx 8 \times 10^7$ cycles, or about 30 μs . That is seven to eight orders of magnitude beyond any measured plasmon coherence time. Coherence is thus not a mild correction to the rate; it is part of the central bound.

6. The Downstream Reaction and Its Signatures

Whatever the capture rate, the neutron it produces has a clean fate. An ultra-low-momentum neutron is captured by ^{10}B in the borophane,



The thermal $^{10}\text{B}(n,\alpha)^7\text{Li}$ reaction proceeds to the excited $^7\text{Li}^*$ state in about 94% of events — each emitting a 478 keV gamma — and to the ground state in the remaining ~6%, so the 478 keV line is an abundant diagnostic. Crucially, the reaction releases its energy as charged particles with no hard primary gamma, which is why the 'missing X-ray' objection levelled at metallic-hydride claims does not apply here. The three observable signatures are excess heat, ^4He , and the 478 keV gamma.

7. What a Detector Would See

Because the rate is bounded (§3), we do not claim a predicted signal that beats noise. Instead we record the minimum surface activity each detector could see, so that the §9 experiment is framed as either a detection or a quantitative limit. Converting noise floors to an equivalent surface power density gives Table 2.

Table 2. The nuclear-product detectors are about 10^7 times more sensitive than calorimetry. A calorimetric signal without accompanying ^4He and 478 keV gammas would be evidence against the mechanism, not for it.

Detector	Noise floor	Min. detectable surface power
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Isothermal calorimeter	$\sim 1 \text{ mW on } 1 \text{ cm}^2$	$\sim 10^{-3} \text{ W} \cdot \text{cm}^{-2}$
Quadrupole mass-spec (^4He)	$\sim 10^9 \text{ atoms} \cdot \text{cm}^{-2}/\text{week}$	$\approx 7 \times 10^{-10} \text{ W} \cdot \text{cm}^{-2}$
HPGe γ -spec (478 keV)	$\sim 10^{-3} \text{ cps, } 5\% \text{ eff.}$	$\approx 1.5 \times 10^{-10} \text{ W} \cdot \text{cm}^{-2}$

The quantitative inputs that would set the rate are listed in Table 3, with honest ranges; the surface field dominates, which is what motivates the §9 scan.

Table 3. Parameter ranges. We omit the earlier Monte-Carlo 'fraction above threshold,' because it is fixed by the assumed field prior and the bounded rate — making it an input, not evidence.

Input	Plausible range	Status / note
Surface field E_{rms}	$10^7 - 10^{11} \text{ V} \cdot \text{m}^{-1}$	Threshold at the extreme upper end; the dominant variable (§9 scan).
η_{loc}	$\sim 0.3 - 1.0$	Geometric decay ≈ 1 ; many-body screening open (§4).
Plasmon lifetime τ	$0.1 - 1 \text{ ps}$	Coherence over $n \approx 8 \times 10^7$ cycles not met (§3, §5.3).
$ \psi(0) ^2$ enhancement	$10 - 10^4$	Needs a DFT calculation (§10).
Adsorbate coverage	$10^{13} - 10^{15} \text{ cm}^{-2}$	Borophane-class 2D hydrides [9].

8. The Standard Objections, Revisited

- **Screening.** The two-dimensional geometry removes the geometric suppression ($\eta_{\text{loc}} \approx 1$); the strong-field, many-body screening of the longitudinal field remains open and should be modelled explicitly.
- **Decoherence.** Real, and — at the N-photon order required — severe. It is part of the §3 bound, not a separate small effect.
- **Absent X-rays.** Resolved by the boron downstream: no hard primary gamma is expected, and the 478 keV line (the $\sim 94\%$ branch) is itself a diagnostic.

9. A Single Experiment That Could Settle It

One sample; the plasmon drive varied across the predicted threshold; all three detectors read out in coincidence. No background produces simultaneous excess heat, ^4He , and 478 keV gammas in the predicted ratio, so the test is robust against calorimetry artifacts and is mechanism-agnostic: a positive result is a threshold turn-on regardless of which mechanism supplies the energy, while a null result bounds the capture rate at these fields — sharpening, or confirming, the theoretical bound of §3. The protocol has four parts.

- **Field-amplitude scan.** Vary the drive across $10^8 - 10^{11} \text{ V} \cdot \text{m}^{-1}$, calibrated with an s-SNOM probe, and look for a sharp turn-on near $10^{11} \text{ V} \cdot \text{m}^{-1}$ with super-linear scaling.
- **Substrate and temperature scan.** Tune the plasmon lifetime through the substrate (free-standing, h-BN, SiO_2) and the temperature (4 K, 77 K, 300 K).
- **Multi-detector coincidence.** Time-correlate calorimetry, ^4He mass spectrometry, and a 478 keV HPGe spectrometer; any one signal without the others is evidence against.
- **Null samples.** Borophane-free graphene, graphene-free borophane, and plasmon-off runs must all register no excess.

10. What This Paper Does and Does Not Claim

It does not claim that low-energy electron capture has been observed, nor that the rate is non-negligible — §3 establishes the opposite, as a rigorous bound. It does claim three things: that the energy-gap objection alone is incomplete, because the dressed-mass threshold is real physics; that the coherent N-quantum rescue does not work, for the specific reasons given in §3; and that the downstream chain provides a unique, artifact-resistant signature that makes the question experimentally decidable. Three calculations would close the case, in priority order: (1) a genuine collective/many-body rate estimate — the only route that could change the bound; (2) a DFT calculation of $|\psi(0)|^2$ at the proton site; and (3) a dielectric-breakdown and material-survival estimate at the required field. None of them changes the falsifiable prediction that the §9 experiment is built to test.

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Appendix A. Threshold and the Dressed Mass

The cycle-averaged Volkov dispersion gives $\tilde{m} = m_e \sqrt{1+\xi^2}$. Requiring $\tilde{m}c^2 \geq (m_n - m_p)c^2 = 1.293 \text{ MeV}$ gives $\sqrt{1+\xi^2} \geq 2.531$, hence $\xi_{\text{th}} = 2.324$.

Appendix B. The N-Quantum Coupling, the Bessel Resummation, and the $n \geq N$ Floor

The effective N-quantum vertex of Eq. (4) is generated at N-th order in the single-quantum coupling $g_0(a^\dagger \sigma_- + \text{h.c.})$, through N–1 virtual intermediate states, giving $g_{\text{eff}} \sim g_0^N / \Delta^{N-1}$ with $\Delta \sim \hbar\omega_0$. Substituting into $\Gamma \propto |g_{\text{eff}}|^2 \langle n^N \rangle$ yields $\Gamma \propto (g_0 \sqrt{n}/\Delta)^{2N} = \xi^{2N}$; resumming all orders gives the generalized Bessel amplitude, $|J_N(\xi)|^2 \approx (2\pi N)^{-1} (e\xi/2N)^{2N}$ for $N \gg \xi$. The n^N coherence bonus is exactly cancelled by the g_0^N coupling. Independently, $a^N|n\rangle = 0$ for $n < N$, so the occupation cannot fall below N; the coherent-rescue result $n^* \rightarrow 1$ is the artefact of an order-independent coupling. At $\xi = 2.3$, $|J_N|^2 \approx 10^{-1.4 \times 10^6}$ for $N = 1.4 \times 10^5$, and $10^{-1.2 \times 10^9}$ for $N = 7.8 \times 10^7$.

Appendix C. The Local-Field Factor η_{loc}

For the 2D eigenmode the matrix element factorises into $F_{\text{height}} = e^{-qd}$, a hydrogenic form factor $[1+(qa/2)^2]^{-2}$, RPA impurity screening $[1+4\pi\alpha_{\text{imp}}q]^{-1}$, and a Debye–Waller factor $e^{-q^2\langle\delta r^2\rangle/2}$. With $q = 10^7 \text{ m}^{-1}$, $d = 1 \text{ \AA}$, $a = 0.5 \text{ \AA}$ the product is $\eta_{\text{loc}} \approx 0.999$; this captures geometric decay only, as noted in §4.